Lists and indexing

A list stores several values in one variable. Write it in square brackets: fruits = ["apple", "banana", "cherry"]. Each item has an index. Indexes start at 0, so fruits[0] is "apple" and fruits[1] is "banana". Negative indexes count from the end: fruits[-1] is "cherry".

Changing a list

append adds an item at the end: fruits.append("mango"). insert(index, value) adds at a position: fruits.insert(0, "kiwi"). remove(value) deletes the first matching value. pop(index) removes by position and returns the item; pop() with no argument removes the last item.

Slicing

fruits[1:3] returns the items at index 1 and 2. The start index is included and the end index is not. Leaving out a number means from the start or to the end, for example fruits[:2] and fruits[1:].

Loops and comprehensions

A for loop visits every item: for fruit in fruits: print(fruit). A list comprehension builds a new list in one line: squares = [x * x for x in range(5)] gives [0, 1, 4, 9, 16].